

MATH 250: TOPOLOGY I EXAM 1

FALL 2026

You have 80 minutes to do these problems in any order. There are 36 points total. Please show work unless otherwise specified.

You are allowed to look at any notes on paper that you wrote prior to the exam. However, you are not allowed to use electronic devices of any kind—including phones, computers, tablets, or audio devices—or any software, including AI tools. If you need to use the bathroom, please give any phones or other electronics to the proctor first.

Problem 1 (6 points). Let d_1 and d_2 be metrics on a set X . Decide whether or not $d(x, y) := d_1(x, y) + d_2(x, y)$ must also be a metric on X . If so, prove it; if not, give a counterexample.

Solution. It is indeed a metric.

- (a) Since $d_1(x, y), d_2(x, y) \geq 0$ for all $x, y \in X$, we have $d_1(x, y) + d_2(x, y) \geq 0$, with equality if and only if $d_1(x, y) = 0$ and $d_2(x, y) = 0$. The latter occur if and only if $x = y$. Therefore, $d(x, y) \geq 0$, with equality if and only if $x = y$.
- (b) For all $x, y \in X$, we have

$$d(x, y) = d_1(x, y) + d_2(x, y) = d_1(y, x) + d_2(y, x) = d(y, x).$$

- (c) For all $x, y, z \in X$, we have

$$\begin{aligned} d(x, z) &= d_1(x, z) + d_2(x, z) \\ &\leq d_1(x, y) + d_1(y, z) + d_2(x, y) + d_2(y, z) \\ &= d_1(x, y) + d_2(x, y) + d_1(y, z) + d_2(y, z) \\ &= d(x, y) + d(y, z). \end{aligned} \quad \square$$

Problem 2 (8 points). Let $\mathcal{B}$ be the collection of rays of the form $(\alpha, \infty) \subseteq \mathbf{R}$, for $\alpha \in \mathbf{R}$. Show that:

- (a) $\mathcal{B}$ is a basis (for some topology).
- (b) $\mathcal{B}$ does not generate the analytic topology on $\mathbf{R}$.

Solution. (a) (I) $\mathbf{R} = \bigcup_{\alpha \in \mathbf{R}} (\alpha, \infty)$. (II) Suppose $(\alpha, \infty), (\beta, \infty) \in \mathcal{B}$. Let $\gamma = \max\{\alpha, \beta\}$. Then $(\gamma, \infty) = (\alpha, \infty) \cap (\beta, \infty) \in \mathcal{B}$.

- (b) Let $\mathcal{T}$ be the topology generated by $\mathcal{B}$. Any element of $\mathcal{T}$ must be a union of elements of $\mathcal{B}$. In particular, if not empty, then it must contain (α, ∞) for at least one real α . So an interval (α, β) with $\alpha < \beta < \infty$ belongs to the analytic topology, but not to $\mathcal{T}$, since it does not contain (α, ∞) . $\square$

Problem 3 (8 points). Let $f: \mathbf{R} \rightarrow \mathbf{Z}$ be the function that sends each real x to the largest integer less than or equal to x . Give $\mathbf{Z}$ the discrete topology. Show that:

- (a) f is continuous with respect to the lower-limit topology on $\mathbf{R}$, generated by the basis $\{[a, b) \mid a < b\}$.

- (b) f is not continuous with respect to the analytic topology on $\mathbf{R}$, generated by the basis $\{(a, b) \mid a < b\}$.

Solution. The key point is that $f^{-1}(n) = [n, n + 1)$ by definition.

- (a) Pick any $S \subseteq \mathbf{Z}$. We must show that $f^{-1}(S) \subseteq \mathbf{R}$ is open in the lower-limit topology $\mathcal{T}_\ell$. Explicitly, $f^{-1}(S) = \bigcup_{n \in S} [n, n + 1)$, so it is a union of elements of a basis for $\mathcal{T}_\ell$. So it belongs to $\mathcal{T}_\ell$.
- (b) Since any singleton $\{n\}$ is open in the discrete topology on $\mathbf{Z}$, it suffices to show that $f^{-1}(n) = [n, n + 1)$ is not open in the analytic topology on $\mathbf{R}$. Indeed, there is no $\epsilon > 0$ such that $(n - \epsilon, n + \epsilon) \subseteq [n, n + 1)$. $\square$

Problem 4 (7 points). Give $\mathbf{R}$ the analytic topology. Let $K = \{\frac{1}{n} \mid n \in \mathbf{Z}_+\} \subseteq \mathbf{R}$. Do at least one of the following:

- (a) Show that $\mathbf{R} - K$ is not open in $\mathbf{R}$.
- (b) Show that the subspace topology that K inherits from $\mathbf{R}$ is its discrete topology.

Solution. (a) Recall that for all $\epsilon > 0$, there is some integer $n \in \mathbf{Z}_+$ such that $\frac{1}{n} < \epsilon$, by the archimedean principle. So $\frac{1}{n} \in (-\epsilon, \epsilon)$, meaning $(-\epsilon, \epsilon)$ cannot be a subset of $\mathbf{R} - K$.

(b) For all $n \in \mathbf{Z}_+$, let $\delta_n = \frac{1}{n} - \frac{1}{n+1}$. Observe that for $n \geq 2$, we have $\delta_n < \frac{1}{n-1} - \frac{1}{n}$. Therefore, $(\frac{1}{n} - \delta_n, \frac{1}{n} + \delta_n) \cap K = \{\frac{1}{n}\}$, meaning this singleton set is open in the subspace topology. So the discrete topology is coarser than the subspace topology, which forces them equal. $\square$

Problem 5 (4 points). Let $f_1: A \rightarrow X_1$ and $f_2: A \rightarrow X_2$ be continuous maps of topological spaces. Show that $f: A \rightarrow X_1 \times X_2$ defined by

$$f(a) = (f_1(a), f_2(a))$$

is continuous with respect to the product topology on $X_1 \times X_2$. *Hint:* Check on a basis.

Solution. Since the sets $U_1 \times U_2$ with $U_i \subseteq X_i$ open for $i = 1, 2$ form a basis for the product topology on $X_1 \times X_2$, it is enough to check that $f^{-1}(U_1 \times U_2)$ is open for all such U_1, U_2 . Indeed,

$$f^{-1}(U_1 \times U_2) = \{a \in A \mid f_1(a) \in U_1, f_2(a) \in U_2\} = f_1^{-1}(U_1) \cap f_2^{-1}(U_2).$$

Since $f_1^{-1}(U_1), f_2^{-1}(U_2)$ are open in A , so is their intersection. $\square$

Problem 6 (3 points). Let $X = \{a, b, c\}$. Write down the coarsest topology on X in which $\{a, b\}$ and $\{b, c\}$ are open. Justify your answer.

Solution. In any topology $\mathcal{T}$ on X where $\{a, b\}$ and $\{b, c\}$ are open, we must have $\emptyset, \{a, b\}, \{b, c\}, X \in \mathcal{T}$. Therefore, we must have $\{b\} = \{a, b\} \cap \{b, c\}$. So it remains to check that $\mathcal{T}_0 = \{\emptyset, \{b\}, \{a, b\}, \{b, c\}, X\}$ really is a topology. Indeed: $\mathcal{T}_0$ contains $\emptyset$ and X , is closed under unions, and is closed under intersections. $\square$